

## **UCCN2243 Internetworking Principles and Practices**

### **UCCN2003 TCP/IP Internetworking**

This group assignment consists of 2 questions. The due date of the assignment is on week 10, Wednesday, 16 April 2025 before 5pm.

This is a group assignment, **group of 3 students MINIMUM OR group of 4 students MAXIMUM.**

Group members **MUST** be from the same Practical LAB session. Marks will be **DEDUCTED** if the assignment group is **LESS** than 3 students (**UNLESS** pre-assigned by the instructor)

The report cover page should include:

- 1) Group member's names,
- 2) Student ID
- 3) Practical group (eg. P1, P2, P3)
- 4) Programme (eg. CN, CT, CS, IA)
- 5) Subject code.

This work should be submitted in softcopy. (Assignment submission link will be provided in WBLE)

***WARNING: You are STRONGLY advised NOT to copy or allow your report to be copied by your friends. Heavy penalty will be enforced on those found to have committed this offence.***

***Sufficient referencing (IEEE style) are required for contents cited from other resources.***

## **Assignment Q1 (10 marks)**

Refer to the network in Q1.pkt

- Company AA network is assigned the public address 201.1.1.1. It has an offsite network which hosts the Remote DHCP Server0 that could be access with the public address 201.1.1.2.
- Configure the remaining gateways with the last usable address.
- Configure Port Address Translation (PAT) on Router0 and Router1 to enable the end devices to access the public network.
- Configure default route on both routers.
- Configure Router0 and Router1 to enable PC0, PC1 and PC2 to access the DHCP service from Server0.

a) Show the running configurations of Router0 and Router1. Highlight the relevant parts and provide brief explanation.

b) Show that PC0, PC1 and PC2 have successfully obtained their IP addresses from the Remote DHCP Server0. Include descriptions.

c) Could PC0, PC1 and PC2 DIRECTLY ping the Remote DHCP Server0? Explain WHY / WHY NOT.

You may use “print screen” to present the results.

## Assignment Q2 (15 marks)

Use GNS3 to build the network as shown in the Figure Q2.

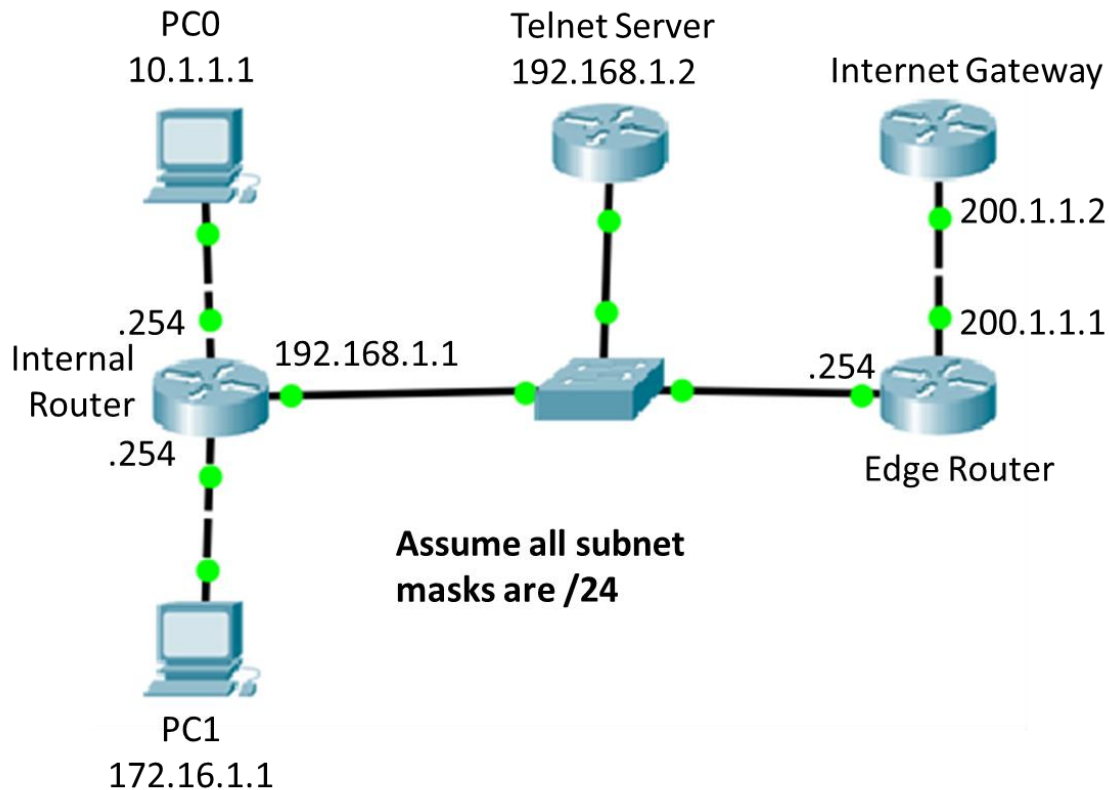


Figure Q2

- Assume static routes are configured for all the routers. Remember to enable telnet service in the Telnet Server.
- PC0, PC1, and Telnet Server are located in their respective Private Networks bounded by the Edge Router with the Public IP of 200.1.1.1
- NAT is configured on the Edge Router to enable the IP range (192.168.1.1 - 192.168.1.254) access to the Internet as well as the public (Internet Gateway) to access the Telnet Server

a) Show (include descriptions) the relevant running configuration for the Edge Router.

b) Show (include descriptions) the relevant running configuration for the Internal Router to enable PC1 to access the Internet. PC0 however could only access the internal devices (PC1, Telnet Server). **Hint: This task should be accomplished using NAT.**

c) Use Wireshark (integrate it to GNS3) and show (include descriptions) the following:

- i) PC1 is able to Ping the Internet Gateway and all other internal devices.
- ii) PC0 is unable to Ping the Internet Gateway but able to Ping all other internal devices.
- iii) The relevant NAT occurring in the Wireshark results

d) Show the Internet Gateway is able to connect to the Telnet Server using telnet, but unable to Ping to the Telnet Server

You may use “print screen” to present the results.

Hints:

GNS3 is an opensource program which you can download for free.

The router’s image file as well as the image file for the end host (you may use the Qemu images) are made available to you. **NOTE: You may use the image files needed for GNS3 from the links provided in WBLE.**

You will need to find out how to integrate Wireshark into GNS3 in order for it to capture and observes the ping packets.